

The Interpretation Session: Program ω 's First Verdicts

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Companion to the gate note (Revision 1.3), which closes on everything measurable. This note is different in kind, not degree: nothing here is two-implemented in the gate note's sense, because nothing here is a computation. It is the session where the corpus's own theorems were asked what they mean, and where two elegant syntheses each survived with exactly one leg honestly relabeled — which is what this project has learned to call knowing something, rather than having guessed well.

1. Four questions, posed formally and plainly

The session opened with four interpretation questions, each pairing a formal statement with its plain-language content:

- **Q1:** Does the expanding flow's monotone, off-spine depth-motion constitute *settling*, given it departs the quasi-free family immediately rather than moving along it?
- **Q2:** What does the three-class taxonomy (60/12/12) mean physically — is anchor-class a property of a possible history, or bookkeeping?
- **Q3:** Does the 24-anchor frozen core earn a “memory” reading, and does it give the anchor-supplier question (REG-13) its first dynamical handle?
- **Q4:** Does the live scalar channel settle the division of labor between dynamics and algebra for measurement stability?

2. Q1: resolved, by catching the question's own stale premise

[FINDING] The tension in Q1 was itself an artifact. “Settling as movement through the family” is $(\mathbb{S}, \tau, \omega)$ -era doctrine, from when χ was the family parameter. Current Foundation demotes the whole equilibrium restriction to [COMMITMENT, not forced]; Becoming's own χ_{kin} is defined pointwise, off-spine, needing no family at all — its own text states plainly that “depth was never actually a thermal notion.” Once this is seen, the flow leaving the family immediately isn't a strike against settling — it's Becoming's own off-spine prediction, realized.

[FACT] The family retains three legitimate roles, and loses exactly one.

It remains the calibration line (depth collapses to one number only at equilibrium, exactly as temperature does); the substrate for the corpus's hardest theorems (disjointness, KMS uniqueness, the type-III verdict are all *about* family members); and the supplier of plausible initial conditions (the flow's own $\nu = 1.5$ start). What it no longer supplies is the rail settling is required to

run on. Discriminator 3 is relabeled accordingly: from a settling criterion to a **spine-adherence diagnostic**, measuring how fast a flow leaves the scaffold, not whether it counts as becoming.

[RESOLVED, §6] The connection flagged above is now settled by the third flow, with one clause corrected in the settling. The $(0, 16)$ twelve do both — deepen cleanly while conserving an exact record of their anchor’s identity. But this proves narrower than the program’s founding demand, not equal to it: the frozen core is **axis-memory, not sign-memory**. The kernel-invariance condition (hence every anchor’s class) is identical under J_0 and $-J_0$ — verified directly, all three tested anchors unchanged — so the dynamics can carry which anchor (which point of the locus, which failure-of-multiplication) but is structurally, provably incapable of carrying which mirror. The mirror bit — the actual founding demand — remains exactly where Paper V’s carrier verdict placed it: with the anchored mechanism class and REG-13, untouched by this taxonomy. Settling can carry content through history; whether it can carry *orientation*, in the sense Phase 4/5 meant it, is answered here as no, by the same sign-blindness theorem this construction was independently shown to satisfy.

3. The three-selections filing

[FACT, verified against source] The corpus has exactly three structurally parallel unsupplied selections: the measurement outcome, the orientation/anchor, and the direction/arrow. Each shares the same shape — the framework derives the alternatives, derives what governs them, proves its own incapacity to select, and locates the selection outside the derivable. Verified precisely: the order-robustness argument (Paper III) genuinely holds at every order, not merely as a stated hope; the gate’s sign-blindness (this note’s companion) is genuine and structural, not a check that happened to pass.

[CORRECTION, first pass] The three are not quite parity in one respect. Measurement carries derived statistics (Gleason-Busch forces the Born-rule form); orientation and direction do not — nothing in the corpus assigns any probability, even a symmetric one, between the two mirror classes or the two ends. The first framing of this (“provably zero derived odds”) overreached: “odds” isn’t a category that applies to a one-time initial condition at all, rather than applying and returning a null value.

[CORRECTION, second pass — F’s, deepening the first] Canonical measures do exist, just none is committed as a sampling law. The wall-null $\rho = \text{Id}/16$ and the Haar average $\langle T_A \rangle = 2 \cdot \text{Id}$ over the anchor locus are both genuine, computed measures over these spaces — the corpus’s own explicit “Haar” language for the second is verified present. What’s missing is any derivation that the world samples from either. Adopting one as a phys-

ical sampling law would itself be a further [COMMITMENT], at the same tier as the equilibrium restriction. **The odds question therefore regresses rather than dissolving or returning zero:** any proposed measure over a selection is itself one more unsupplied selection, one level up.

[INTERPRETATION, flagged for pressure-testing rather than endorsed outright] The three-way parity with established open problems in physics (chirality as unexplained input; the measurement-selection gap; now the cosmological measure problem, in its textbook shape) is a striking pattern. Given how many syntheses this session has needed to correct on inspection, this is recorded as worth stating once, precisely, and testing hard — not yet adopted as a settled frame.

4. The line-crossing argument: assembled, then corrected twice at the same joint

[ARGUMENT, as first assembled] The disjunction is derivable even where the disjunct is not. Three pieces: the label space is binary (derived); every realized world has a sharp label (claimed via factoriality, $M_\omega = \mathbb{C}1$); nothing derivable supplies which value (the protection theorems). Read together: the framework would prove a selection occurred, prove it binary, and prove it cannot say which way — “the unknowable enters carrying a proof of its own arrival.”

[ERROR, caught and corrected] The middle leg cited the wrong axis. Factoriality ($M_{\omega_\beta} = \mathbb{C}1$) is a proven fact about the β -axis — trivial center within one thermal state’s own algebra. It says nothing about orientation or anchor. What the argument actually needed — an analogous disjointness result across different orientations or anchors — does not exist in the corpus; searched for directly and found nothing.

[FACT, F’s resolution, verified] The gap is not merely open — it is closed, false, in the finite construction, with the true axis named. The wall sector is explicitly finite-mode (confirmed in Phase 5’s own text). On any finite-mode CCR/CAR system, the Shale–Stinespring/Powers–Størmer criterion (quasi-equivalence iff the symbol-difference is Hilbert–Schmidt) is satisfied automatically, since every operator on a finite-dimensional space is trivially Hilbert–Schmidt. **The four orientation vacua are therefore genuinely quasi-equivalent as built — superpositions between them are legal states, not merely unforbidden.** This was true the moment the construction was fixed as finite-mode, before anyone checked it.

[REGISTRY ENTRY, filed] REG-15 — the sharp-realization question. Does the framework’s own construction force a physically realized world to carry a definite orientation/anchor value? **Resolved NO, as a theorem, in any finite truncation** (Shale–Stinespring triviality, exact). **Open in the**

thermodynamic limit, where the deciding criterion is named: orientation-sectors become disjoint exactly when the mirror-flip acts with positive density across infinitely many modes. The corpus's existing infinite-volume constructions live on the bosonic scale line; the fermionic/wall module has not been taken to its own limit — this is the actual, located open work.

[INTERPRETATION, F's reframing, held at its earned weight] If REG-15 resolves YES in the limit, folium-tier sharpness joins type III, event-coarseness, and thermal disjointness as an infinite-volume phenomenon — the world's definiteness about which mirror it is would share its source with its indefiniteness about individual events. A genuinely elegant possible unification, explicitly not yet a theorem.

5. The ledger, this session

Species named, both recurring from prior arcs and new: - *Theorem-scope inflation*, second occurrence (the λ^3 -split's fixed- W scope; now factoriality's β -axis scope). Countermeasure strengthened per the recurrence: any load-bearing theorem citation in a synthesis quotes the theorem's own statement inline, so an axis mismatch is visible in the same line it's used. - *The elegance tax* (new): the most beautiful synthesis in an exchange reliably carries exactly one seam needing exactly this kind of check — named as an observed regularity, not a criticism of the thinking that produces it. - One precision correction on the citation register itself (the wall-null/four-vacua connection stated as a well-supported inference rather than a verbatim quote, before it entered this note as settled).

6. The third flow: pre-registration confirmed, the taxonomy completes

[FACT, two-implemented] The pre-registered prediction landed exactly, confirmed independently by F, digit-for-digit. The $(0, 16)$ pure-antilinear anchor ($e_1 + e_{12}$, $\text{tr } A = +16$) was integrated from the same quasi-free initial state. Result, checked branch by branch across the full trajectory, not sampled: all 8 distinct symplectic eigenvalues are monotone nondecreasing throughout, no exception. The 4-dimensional frozen core stays exactly at $\nu = 1.5$ (deviation 0.0); every active branch strictly increases, matching $2e^6 \approx 806.86$ and $2e^{12} \approx 325509.6$ within integration tolerance — explaining, not merely reproducing, the two growth rates. The falsifier — any decreasing active branch — did not fire.

This is deepening with memory, the third character, completing the taxonomy:

Class	Anchor	Frozen core	Behavior	Character
Generic (4, 12), 60 members	$e_1 + e_{10}$	No	Deepens	Deepen-and-forget
Extreme-low (12, 4), 12 members	$e_4 + e_9$	Yes	Wanders	Remember-and-wander
Extreme-high (0, 16), 12 members	$e_1 + e_{12}$	Yes	Deepens with memory	Deepen-and-remember

The fourth logically available combination — wander-without-a-core — is structurally absent: the frozen core is provably an extreme-class-only property, so no generic anchor can carry one.

[INTERPRETATION, dynamically supported] Q3’s promotion: the frozen core survived its first genuine dynamical test — held at exact stasis through five orders of magnitude of surrounding growth — supporting “the frozen core is the anchor’s conserved record” as an adopted reading, not merely a static observation. REG-13 gains the refinement that any anchor-supplier now faces a typed choice among anchor classes, not just a point on the locus — though, per the axis/sign correction above, that typed choice concerns *which anchor*, not *which mirror*.

Two items remain, side by side, neither privileged: REG-15’s thermodynamic-limit half, and the sign-carrier’s supplier (Paper V’s carrier verdict, REG-13) — both mathematics still ahead. Q2 is answered empirically by the taxonomy above: the classes are not bookkeeping but three distinct relationships between history and memory. Q4 stands as drafted, pending explicit adoption.

Created in collaboration with Claude (Anthropic). Interpretation-session record, distinct in kind from the gate note’s computational register; all conclusions and any errors remain the author’s responsibility.